Year 12 Further Maths - Geometric distributions

Similar questions - Sampling, interviews and cars

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Each of three students, X, Y and Z, is given an identical pack of 40 cards, of which 10 are red and 30 are blue. They are each told to carry out a different experiment.

- Student X: choose a card at random 15 times altogether **with** replacement, and record the number of red cards obtained.
- Student Y: choose a card at random 15 times altogether **without** replacement, and record the number of red cards obtained.
- Student Z: choose single cards at random **with** replacement until the first red card is obtained, and record the number of cards chosen including the first red card.

(a) Find the probability that student Z has to choose more than 7 cards. **[3]**

(b) Each student carries out their experiment many times. State, giving a reason, which student's results you would expect to have (i) the largest variance, (ii) the smallest variance. **[4]**

(c) Find the expected mean and variance of the results of student X, and the expected mean of the results of student Y. **[3]**

2 A town council is planning to introduce a new set of parking regulations. An interviewer contacts randomly chosen people in the town and asks whether they are in favour of the proposal. The first person who is not in favour is the Rth person interviewed. It can be assumed that the probability that any randomly chosen person is not in favour is a constant p, where $0 < p < 1$. Assume first that $\mathrm{E}(R) = 8$.

(a) Determine $\mathrm{P}(R \geqslant 11)$. **[3]**

Now, without the assumption that $\mathrm{E}(R) = 8$, consider a general value of p. It is given that $\mathrm{P}(R = 3) - 0.5\,\mathrm{P}(R = 2) - a\,\mathrm{P}(R = 1) = 0$, where a is a positive constant.

(b) Determine the range of possible values of a. **[7]**

3 Anika walks along a street that contains parked cars. The number of cars that Anika passes, up to and including the first white car, is denoted by X.

(a) State two assumptions needed for X to be well modelled by a geometric distribution. **[2]**

Assume now that $X \sim \mathrm{Geo}(p)$, where $0 < p < 1$.

(b) For $p = 0.15$, find $\mathrm{P}(X > 5)$. **[2]**

The number of cars that Anika passes up to but not including the first white car is denoted by Y.

(c) For a general value of p, determine a simplified expression for $\mathrm{E}(Y) + \mathrm{Var}(Y)$ in terms of p. **[3]**

Ben walks along a different street. The number of cars that Ben passes, up to and including the first white car *whose number plate ends in an even digit*, is denoted by Z. It may be assumed that $Z \sim \mathrm{Geo}\left(\frac{1}{2}p\right)$, where p is the parameter of the distribution of X. It is given that $\mathrm{P}(Z = 4) = k\,\mathrm{P}(Z = 2)$, where k is a positive constant.

(d) Determine the range of possible values of k. **[5]**

Question 1

(a) $Z \sim \text{Geo}(0.25)$: $\text{P}(Z > 7) = 0.75^7 = 0.133$

(b) (i) Z: variance $\dfrac{0.75}{0.25^2} = 12$, much larger than the binomial/hypergeometric variances, and is unbounded above. (ii) Y: sampling without replacement removes the independence and reduces the spread: variance $15(0.25)(0.75)\dfrac{25}{39} = 1.80$ compared with 2.81 for X

(c) $X \sim \text{B}(15, 0.25)$: mean 3.75, variance $15(0.25)(0.75) = 2.8125$. Y has the same expected mean $15 \times \frac{1}{4} = 3.75$

Question 2

(a) $\text{E}(R) = \dfrac{1}{p} = 8$, so $p = \frac{1}{8}$ and $\text{P}(R \geqslant 11) = \left(\frac{7}{8}\right)^{10} = 0.263$

(b) Let $q = 1 - p$. $q^2 p - 0.5qp - ap = 0$, so $a = q^2 - 0.5q = q(q - 0.5)$. As $a > 0$ we need $q > 0.5$; and $q < 1$. Since a is increasing on $0.5 < q < 1$ (the vertex is at $q = 0.25$), a takes values from 0 to $q(q - 0.5) \to 0.5$. So $0 < a < 0.5$

Question 3

(a) Cars are white independently of each other; the probability that a car is white is constant

(b) $\text{P}(X > 5) = 0.85^5 = 0.444$

(c) $Y = X - 1$, so $\text{E}(Y) = \dfrac{1}{p} - 1 = \dfrac{1-p}{p}$ and $\text{Var}(Y) = \dfrac{1-p}{p^2}$. Sum $= \dfrac{(1-p)(p+1)}{p^2} = \dfrac{1-p^2}{p^2}$

(d) $\text{P}(Z = 4) = \left(1 - \frac{p}{2}\right)^3 \frac{p}{2}$ and $\text{P}(Z = 2) = \left(1 - \frac{p}{2}\right) \frac{p}{2}$. So $k = \left(1 - \frac{p}{2}\right)^2$. As $0 < p < 1$, $\frac{1}{2} < 1 - \frac{p}{2} < 1$ so $\frac{1}{4} < k < 1$